1 Graphic order sequencing

Overview

Menu	Production control → Production support → Graphic order sequencing
Transaction code	graps
Function authorization	graps

Purpose

The Graphic order sequencing (GAV) is an integrated planning module. Within the scope of the production control, you can use the Graphic order sequencing to generate sequencing lists for an organizational unit. The Graphic order sequencing (GAV) includes the following functions:

- Display of operations that are scheduled for a workplace or still in the pool of groups.
- Planning of an operation for a workplace/machine
- Re-plan an operation on another workplace or group
- Define a processing sequence ("sequencing") that specifies how the operations to be produced are displayed in the sequencing list on the terminal
- Splitting of operations (requires separate license)

The Graphic order sequencing is a cost-effective alternative to the HYDRA shop floor scheduling (HLS) to schedule production orders and sort them in a queue.

Integration

You can use the *Graphic order sequencing* to assign operations to the machines and workplaces of the shop floor.

If you schedule an operation for a workplace, you define the processing sequence in production. The sequencing list of the shop floor terminal shows this processing sequence.

The Graphic order sequencing does not compare the planning with the defined shift model of the workplace as does the HYDRA shop floor scheduling (HLS). The Graphic order sequencing also does not include secondary resources in the planning (production resources and tools, material components, persons).

Requirements

If you want to display the current planning in the graphic order sequencing, the following is required:

- Create orders relevant for planning (production orders, maintenance orders, project orders) in the system or transfer these orders from a higher system.
- Coordinate responsibilities within your company and specify the persons responsible for order planning. Also specify which workplace/machine groups should be planned. Based on these initial steps, you have defined the necessary planning profiles.

Note: You cannot use the *Workplace assignment* of the Personnel Scheduling product group (PEP) and the *Graphic order sequencing* simultaneously. Inconsistencies might occur if personnel and capacities are planned at the same time, as the *Workplace assignment* uses data structures of the *Graphic order sequencing*.

1.1 Graphic order sequencing

The basic element of the *Graphic order sequencing* is the graphic planning board. The graphic planning board is modeled after a traditional planning board.

If you request data in the graphic order sequencing, the planning situation is displayed that corresponds to the currently available state in the database. All order types are relevant for planning that have been configured using the ID "Planning". You can customize these Order types in the system.

The display includes all workplaces/machines or groups for which the planner is authorized (it is checked if the workplace is included in the responsibility area). Optionally, the user can define so-called user profiles. The user can then specify in detail the respective display within the responsibility areas the user is authorized for.

The following illustration shows the application *Graphic order sequencing*:

Note: This screenshot shows *one* possible presentation of the graphic order sequencing. Depending on the settings, the presentation may vary.

In the *Graphic order sequencing*, you can realize an interactive planning by manually changing the current planning situation. You can make the following interactive changes of the planning state:

- Schedule an operation for a workplace

- Re-plan a planned operation

- Reassign an operation to another machine within the machine group

- Deallocate an operation to the pool of groups

Planning changes are stored in the database on saving. The current planning state is discarded, if you exit the Graphic order sequencing without saving the planning.

1.1.1 Toolbar

This chapter describes the functions that you can call via the toolbar. The access to specific functions depends on the assignment of function authorizations.

1.1.1.1 Tab *Main page*

Category *Data*

Data is requested

Data matching the selection criteria entered in the tab *Selection* is read from the database and loaded into the graphic order sequencing.

Cancel

Interrupts the process of requesting data.

Save planning

Function authorization: grapt.save
The current assignment is saved. Changes are stored in the database and are thus available for other planners.

On saving, the internal fields used to sort the terminal's sequencing list are set to the start date and the start time that have been specified by the planner in the graphic order sequencing.

A pop-up dialog appears to notify the planner that changes have been saved.

Category *Options*

Search operation

All operations matching the criteria entered are selected.

Notes on the individual fields:

- Operation: This is a combined order/operation number (MES order number).
- Article: The article/item stored in the operation (see *Order information* > *Operations* > tab *Operation* > field *Article*)

- Tool: The (main) tool stored in the operation (see *Order information* > *Operations* > tab *Operations* > field *Tool*)

You can also run a search using wildcards (at the beginning and/or end of a character string).

Next time you call the function, the dialog shows the selection criteria that you have entered the last time.

Settings

In the dialog *Settings*, you can make different settings, e.g. change the color of bars. The section *Settings* describes the individual options.

Save

You must save changes, if the changed configurations should be available after having closed the application. So save the settings if you perform one of the activities mentioned in the following. Click the *Save* icon in the toolbar to save the changed application layout.

- You add or delete a detail application of the tab *Planning details*.
- You add or delete columns in a table.
- You change the layout of categories and columns in a table.
- Make settings in the dialog *Settings*.

1.1.1.2 Tab *Selection*

Category *Data*

The functions are similar to the ones in Tab Main *page*.

Category *Selection*

planning profile

Planners with many responsibility areas should use planning profiles.

Advantages of planning profiles:

- Better overview
- Faster load times when requesting data

Select a planning profile before requesting data. Using this planning profile, the groups of workplaces/machines that are assigned to the planning profile are loaded and displayed and can then be scheduled.

If you do not enter a planning profile, *all* workplaces are loaded that are assigned to your responsibility areas.

Planning horizon

You can change the dates "Planning horizon from/to" manually. The data selection always refers to the dates entered in the input fields. The system selects all operations with a scheduled start time within this period.

Order type

If you select one or several order types, only those operations are loaded and displayed that belong to an order of the respective order type.

Using this selection option, you can for example select and plan capacity or simulation orders. The planning of the production orders is not affected in this case. Important: Always include all order types when you select production orders before you perform and save a real planning.

1.1.1.3 Tab *View*

Category *Planning board*

Expand/collapse all groups

Open or close the workplace/machine groups.

Zoom in

You modify the Gantt presentation in order to increase the degree of accuracy of the time scale. You can also enlarge the presentation using the key combination Ctrl + "+" (numeric keypad) or using the mouse wheel: turn the mouse wheel while keeping the Ctrl key held down.

Zoom out

You modify the Gantt presentation in order to get a better overview. You can also reduce the size of the presentation using the key combination Ctrl + "-" (numeric keypad) or using the mouse wheel: turn the mouse wheel while keeping the Ctrl key held down.

General view

The general view is an additional window showing the complete Gantt chart. A frame indicates which chart section is currently displayed in the main window. If you move the frame with the mouse, the section displayed in the main window is moved accordingly when you release the mouse button. Similarly, you can zoom the frame. The display detail in the main window is then changed accordingly. And vice versa, size and position of the frame also change, if you zoom or scroll the section displayed in the main window.

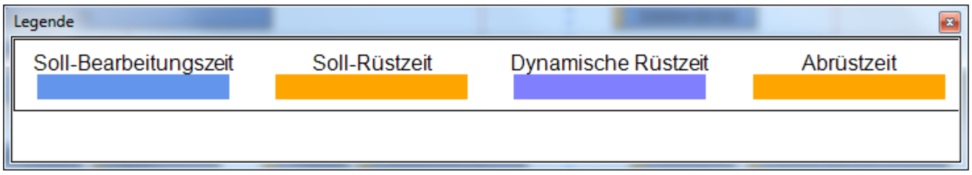
Open and close the *General view* by clicking the relevant button. Every time you request data, the General view is automatically closed.

Legend

The legend view is an additional window to present a legend on the screen. The following information is shown:

- Target processing time (in some cases, shown multiple times because of different color codes!). It is the remaining run time (calculated using the remaining run time formula).
- Target setup time
- Teardown/retooling time

The configurations in the Settings, section Bar layout, specify the respective display.



Open and close the legend by clicking the relevant button. Every time you request data, the legend is automatically closed.

Print preview

The Print preview shows the complete print area. You can restrict or zoom the print area in this view. Refer to the chapter 1.1.7 *Print* preview for more information.

Category *Operations*

Links

Not used

Notes on operations

Function authorization: edopnote

Shows the detail application Notes on operations. The use of this function is described here.

Open and close the detail application by clicking the relevant button.

Show delays

You can identify delayed operations. The use of this function is described here.

1.1.1.4 Tab *Planning details*

pool of groups

Function authorization: grapv.tabgrp

The function call *pool of groups* opens a detail application showing the operations that are planned for a group. The display includes the scheduled operations for *all* groups.

Open and close the detail application by clicking the relevant button.

Available workplaces/Workplace backlog

Function authorization: grapv.tabwp

The function call *Available workplaces/Workplace backlog* opens a detail application showing all scheduled operations of a workplace in tabular form.

Open and close the detail application by clicking the relevant button.

Order network (requires authorization key)

Function authorization: grapv.orview

The function call *Order network* opens an application showing the Order network the selected operation is assigned to.

Open the detail application by clicking the relevant button. Unlike the other detail applications, the order network is not closed by clicking the button again, but by closing the window.

1.1.2 Time scale

A blue, vertical line identifies the current point in time "now". In addition, red vertical lines visualize the end of each day.

The dimension of the time scale is defined in the Settings.

To change the resolution, resize the time scale (by holding down the left mouse button).

Every time you request data, the time scale is set back to its original setting.

The format of the date values displayed on the Gantt chart depends on the format specified by the operating system. The MOC format is not relevant.

1.1.3 Presentation of groups and workplaces/machines

The workplaces/machines that are available for planning are displayed on the left hand side. A workplace is shown if the following criteria are met:

- The identifier *Planning function* is set to 'P' or 'H' in the Workplace and resource configuration.
- The workplace is not "blocked" in the *Workplace and resource configuration.*
- The workplace is assigned to a Group that is defined as a capacity group.
- The group the workplace is assigned to is also assigned to the selected planning profile and configured there as *Visible in shop floor planning*.
- The workplace/machine is part of the planner's Responsibility area.

If you click Request data, the overview of the workplace groups is shown. The group number is shown. You can define the order of the groups using the field *Order* in the configuration of the *planning profile*.

Note: The values in the field *Order* must be unambiguous within a planning profile. Otherwise the order is random.

Select a group by clicking it. The workplaces/machines of this group are shown. Use the field *Position* in the *Group assignment* configuration to specify the order in which workplaces of a group are displayed. Note: The values in the field *Position* must be unambiguous within a group. Otherwise the order is random.

List of workplaces: Columns

The following columns are available for display in the list of workplaces:

Header	Meaning	Comment
(none)	Status color or RPA color (according to the Setting)	Due to the linked logic, group workplaces always have the status *Production* (assigned to the resource performance account *Main utilization*). You can define the update interval of the status in tab *General* of the settings.

Header	Meaning	Comment
Workplace	Number of the workplace/machine	
Short name	Short name of the workplace/machine	
Designation	Full name of the workplace/machine.	
Group	Group of the workplace/machine.	
Symbol	Graphic	
Performance level	Current performance level of the workplace/machine.	The performance level is read from the database when data is requested.

The planner defines in the tab *General* of the settings which columns are displayed in the table. The configuration is saved user-specifically. Change the column widths directly in the table (click and drag across). This modification is saved user-specifically.

Tooltip

If the mouse pointer rests on a workplace, a tooltip is displayed providing the following data:

Header	Meaning	Single workplace	Group workplace
Workplace	Number of the workplace/machine according to configuration	X	X
Short name	Short name of the workplace/machine according to configuration	X	X
Designation	Full name of the workplace/machine as configured	X	X
Cost center	Cost center of the workplace/machine as configured	X	X
Workplace type	Workplace type as configured: E = Individual workplace G = Group workplace	X	X
Status	Status number and name of the workplace/machine status as configured	X	-
Status since	Point in time (date, time) since when the status has been active.	X	-

1.1.4 Functions on the level of the group/workplace

You can perform the functions described in the following if you select a workplace or group in the column *Workplace* and right-click it (to open the context menu). The order in the context menu may vary from the order the functions are described here:

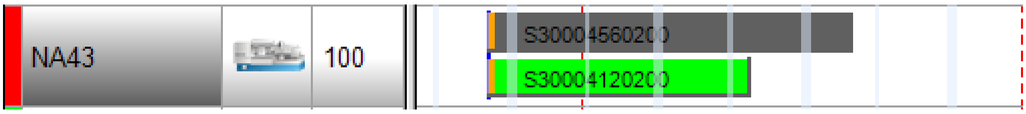
Workplaces/Machines

The menu item *Workplaces/machines* calls the application *Workplaces/Machines* for the current workplace/machine or the current group. The workplace number or group is transferred to the application as selection criterion.

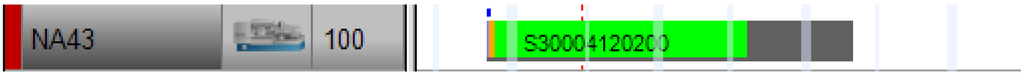
Single-line presentation

With single-line presentation, operations assigned twice are not displayed one below the other (i.e. listed) but one on top of the other (i.e. overlapping).

Two-line presentation:



Single-line presentation:



- With single-line presentation, the label text on OP bars might partly not be visible.
- It is only a temporary presentation option. Every time you request data, parallel operations are again displayed one below the other.

Selecting operations in the pool of groups

This entry is available if you call the context menu on group level. Click this entry to select all operations included in the pool of groups.

Select the planned operations

The entry is available if you request the context menu on workplace level. Click this entry to select all operations planned for this workplace.

With the functions mentioned above, you can select multiple operations by holding down the CTRL key and clicking further operations.

1.1.5 Displaying operations in the Gantt

Order backlog / pool of groups / pool of orders

Operations transferred from the ERP system are generally planned for a group and therefore included in the pool of groups. You can show the pool of groups in tabular form and/or as Gantt chart.

If the machine/capacity group is changed for a workplace, you must **also** change the group for the planned operations in the order backlog. Otherwise, the Graphic order sequencing no longer shows the operations.

Pool of groups in Gantt

In order to graphically display operations in the pool of groups of the Gantt, you must activate the option *Show operations in pool of groups of the Gantt* in the settings in tab *General*. In this case, operations are displayed according to their scheduled start time.

Planned operations

Planned operations are displayed for the individual capacity (workplace/machine), i.e. in the relevant row (only if the view is expanded).

Duration/length of an operation bar

The length of an operation represents the sum total of setup time, processing time/Remaining run time and retooling time (teardown time). This time is called execution time. You can define the process times either directly or the system can calculate these times using a formula.

Setup time is generally integrated in the calculation of execution times in *Graphic order sequencing,* also if the operation has already been started and is currently in status *Interrupted*.

You can customize the configuration and define if the setup time of interrupted operations is integrated in the graphic order sequencing or not. To configure the order type, the following options are available in the field *Setup time in shop floor planning*:

- **R: Remaining setup time**
  Integrate remaining setup time: target setup time minus "times posted to RPA 7"
- **B: Target setup time 0, if OP is started**
  Ignore setup time if the operation has already been started (even if OP is interrupted)

Running operations do generally not include the setup time.

If the current remaining run time of an operation is 0 or negative (e.g. in case of overproduction), the remaining runtime is set to 0 for a running operation. In case of a prepared or interrupted operation, the remaining run time is set to 900 seconds = 15 minutes by default. This way, this operation can still be re-planned manually using the drag & drop function in the Gantt. You can change this duration in the settings of the graphic order sequencing in tab *Bar layout*.

Color coding of operations

You can define the color of operation bars in the settings in tab *Bar layout*.

Exceeding basic dates

Make the following setting to enable the colored presentation of the operation bar:

- Settings → tab *Bar layout*
- Enable the option *Operation delayed.*

If an operation is planned outside the predefined basic dates and the planned end date is later than the "latest end time" (LET), the presentation changes. The operation bar is hatched or displayed in a different color.

Double clicking an operation bar

If the user has the authorization *grapt.edop.edit*, the user can double-click an operation in the Gantt (only operations that are not logged on). A dialog opens and the user can change default values/target data. The following data can be changed in the dialog:

- Target quantity (primary quantity unit)
- Partitioning
- Setup time
- Target cycle
- Teardown/retooling time
- Wait time

Confirm the dialog to store the changed data directly in the operation and to update the graphic order sequencing.

1.1.6 Operation functions

Right-click a selected operation to open a context menu including the following functions:

Deallocate

Use this function to deallocate selected operations. The operations are then available in the pool of groups.

Order overview

The Order overview is opened for the selected operation. The order number of the selected operation is transferred as selection.

Order information

The Order information including relevant default data and current information about the order is opened for the selected operation.

Order network

This menu item opens the application *Order network* for the selected operation. The order network shows the operations that are related to the selected operation in a graphic (Gantt chart).

This function requires the respective authorization (license).

Split operation

Subject to the setting of the option *Enhanced split function* in the Basic settings, you can find the requirements and more information about operation splitting in one of the below-mentioned documents:

- Enhanced split function is not enabled: Operation Split
- Enhanced split function is enabled: Enhanced split function

This function requires the respective authorization (license).

dissolve split

Use the function *Cancel split* to undo the splitting of an operation.

Subject to the setting of the option *Enhanced split function* in the Basic settings, you can find the requirements and more information about operation splitting in one of the below-mentioned documents:

- Enhanced split function is not enabled: Operation Split
- Enhanced split function is enabled: Enhanced split function

Create note

You can create one or more notes for the selected operation. You must have activated the respective detail view Notes to be able to show notes.

Select all operations of the order

The function *Select all operations of the order* selects all operations included in the Gantt chart and belonging to the same order as the current operation.

In connection with the function *Deallocate*, you can remove an entire order from planning. Note: Fixed operations must first be unfixed.

Select this OP and all following OPs of the order

Use the function *Select this OP and all following OPs of the order* to select this operation and all operations included in the Gantt chart that belong to the same order and have a higher operation number than the current operation.

Select the planned operations

Use the function *Select the planned operations* to select all operations scheduled in the Gantt chart.

In connection with the function *Deallocate*, you can remove all operations from planning. Note: Fixed operations must first be unfixed.

1.1.7 Print preview/Print

The functions and options described in this chapter are part of the used Gantt visualization component. As the Gantt visualization component is a standard third-party product, changes to the layout and the functionality are only possible to a limited extent.

You can print the graphic order sequencing using the print preview screen that is opened via the button  in tab *[Different screens](#)* of the toolbar.

You can view each single page or see an overview of all pages of your presentation. You can also interactively zoom in on a section of your chart and then print it.

The status line provides information about the total number of pages and the horizontal and vertical distribution of pages. In the view *Show single page*, the current page number is additionally shown.

Functions of the print preview

Close

You exit the print preview and return to the application.

Note: The settings made in the print preview are currently not saved.

<

Only active if the button *Show single page* has been clicked. If the chart takes up several pages, you can view each single page. Click this button to return to the previous page. You move through the pages from right to left in ascending rows.

>

Only active if the button *Show single page* has been clicked. If the chart takes up several pages, you can view each single page. Click this button to go on to the next page. You move through the pages from left to right in descending rows.

Overview

If the chart takes up several pages, you can either view each single page separately or in the overview. In the *Overview,* all pages are displayed - depending on the number of pages, the view is zoomed out more or less; in the *Single page* presentation mode, initially the first page of the chart is shown and zoomed in. Click [<] or [>] to scroll through the pages. Double-click a page to easily switch between the two presentation types *Single page* and *Overview*.

In the presentation mode *Single page*, you can interactively enlarge specific sections of your chart. To do so, hold down the left mouse button and make a square frame around the section you want to zoom in. As soon as you release the left mouse button, the framed-in image section is enlarged. The label text of the button *Print* changes to *Print area.* You can now print the image section in the currently enlarged size. Note: The zoom factor selected in the print preview does not change the scaling factor in the dialog field *Page setup*.

Fit to single page

Use this button to reduce a chart of several pages to one page. Here, you also have the possibility to interactively enlarge sections of the chart as described in *Single page/ overview* and to print them.

Zoom factor

You can use a zoom factor from the list to change the presentation size of your chart in the print preview. You can only select the zoom factor if single pages are shown. The print is not affected. Depending on the selected size, vertical and/or horizontal scroll bars are displayed. You can also use the mouse wheel to move the image (without shift key vertically, with shift key horizontally).

Auto is preset as zoom factor. With this setting, the size of the page is always reduced or enlarged so that it fills the screen.

Page setup

If you click this button, you are forwarded to the dialog *Page setup*, where you can change the page layout. The options of this dialog are described below.

Print/print area

If you click this button, you are directed to the Windows dialog *Print*. If you have zoomed in a section in the print preview, the label text of the button changes to *Print area*. If you click this button, the option *Location* is preselected in the Windows dialog *Print*. Click *OK* to print the section shown on the screen.

Note: The zoom factor selected in the print preview does not change the scaling factor in the dialog field *Page setup*.

Functions of the dialog *Page setup*

Mode

Select the scaling type from the drop down list and the relevant values for *Zoom factor* or *Maximum width/height* to define the scale of the presentation's output. If you click *Apply* , the values resulting from your settings are displayed in the section *Current***.**

Zoom factor

If the scaling factor is 100%, the original size is shown. A smaller value will reduce the presentation, a larger value will increase it.

Fit to page count (Adjust to the number of pages)

Select this option to define the maximum number of pages for the width and the height. The chart output is then divided accordingly (**maximum width, maximum height**). The charts are presented as large as possible, but they are not distorted.

Zoom with horizontal fit

Select a zoom factor and a fixed number of printed pages (width) that should be maintained by compressing or stretching the time scale during printing.

Repeat title/ table/ time scale/ legend

Check this option to include title, table, time scale and legend on each page of the printed chart.

Show table

Define whether or not to print the table. If this option is not selected, the table is not printed.

Table columns

Here, you define which table columns are printed. Enter the required columns separately or in ranges separated by a comma or semicolon. Example: "1;5-7;3" selects the columns 1, 3 and 5 to 7.

Show diagram

Here, you enable or disable the printing of the chart (time scale and bars).

Adjust time scale to width of pages

This option optimizes the space used on the printed pages:

- If a fixed number of pages is defined for scaling: The zoom factor is calculated so that the full height of the set number of pages is printed. At the same time, the time scale is compressed or stretched so that the full width of the defined number of pages is made use of.
- If a zoom factor is defined for scaling: The time scale is compressed or stretched so that the full width of the defined number of pages is made use of.

Pad pages with space

Use this option to define whether you want to leave space between the chart and the boxes for title and legend. If you enable this option, the boxes are always positioned on the page margin and printed in full width on each page. If this option is disabled, the boxes are printed without space next to the chart. Depending on the chart, the width of the boxes may vary on the different printed pages.

Show frame outside

Activate this checkbox so that a frame is printed around the chart. If the option *Repeat title/ table/ time scale* is enabled, a frame will be printed on each page; otherwise a frame will be drawn around the entire chart.

Alignment

Define the alignment of the charts on the page.

Show crop marks

If you enable this option, crop marks are set on the chart. This way it is easier to glue the printed single pages to create a full chart.

Show folding marks (DIN 824)

For construction diagrams, the German national standard DIN 824 defines a very specific folding procedure that allows you to fold up the drawing to a DIN A4 format size. If this option is enabled, the folding marks on your diagram facilitate the folding procedure. The following formats are available:

- **Form A**: With a filing margin on the left side so that holes can be punched into the folded drawing. The drawing can then be filed in a folder without filing clip.
- **Form B**: This format is somewhat narrower. A filing clip is attached and the drawing plus clip have the format DIN A4.
- **Form C**: The folded drawing is not punched, but placed in a clear cover.

The folding marks can be added for any target format, whereas DIN 824 only includes the formats DIN A0 to A3.

Page numbering

If you check this option, the page number is printed on the bottom left of every page. You can select from the following options:

- **Row.column**: This option is useful if the chart covers more than one page in length and width. The position of the page in vertical order is printed before the dot, the position in horizontal order is printed after the dot.

- **Column.row**: This option is useful if the chart covers more than one page in length and width. The position of the page in horizontal order is printed before the dot, the position in vertical order is printed after the dot.

- **Page/count (total number)**: The first number shows the current page number, the second the total number of pages: 1/6, 2/6, etc.

Text

Enable this option, if you want to add a random text at the bottom left of every page. In some cases, this text is printed to the right of the page number.

For page numbering, you can enter the following placeholders in the row *Additional text*. These placeholders are then replaced with the relevant content when the document is printed.

{PAGE} = Consecutive page number

{NUMPAGES} = Total number of pages

{ROW} = Row position of the detail section in the total chart

{COLUMN} = Column position of the detail section in the total chart

Print date

If you check this option, the print date is printed on the bottom left of every page. In some cases, the print date is printed to the right of the page number and the additional text.

Sheet margins

Define the sheet margins in cm using the fields *Top*, *Bottom*, *Left* and *Right.* You cannot go below the minimum margins that printers preset for technical reasons.

1.2 Settings

If you click , you call a dialog where you can make custom settings. The settings are saved user-specifically. The next time you call the graphic planning, the settings are available.

Some changes are only activated, when you request data the next time. We therefore recommend to request data in the graphic order sequencing after having performed changes.

1.2.1 Tab *Main page*

General settings

Update statuses

The setting *Update statuses* defines if and at which interval the workplace/machine status is read. The updated status is displayed on the left hand side of the *Graphic order sequencing* (column without caption).

Update data

The setting *Update data* defines if and at which interval the *Graphic order sequencing* is updated (i.e. data is read and displayed). If changes have been made in the meantime, a prompt pops up asking whether the changed data should be saved or not.

Update order overview automatically when selecting an operation

The function is available as of Service Pack 16.

If you click on one of the following elements, the order number of the operation is copied into the Order overview and the display in the order overview is updated:

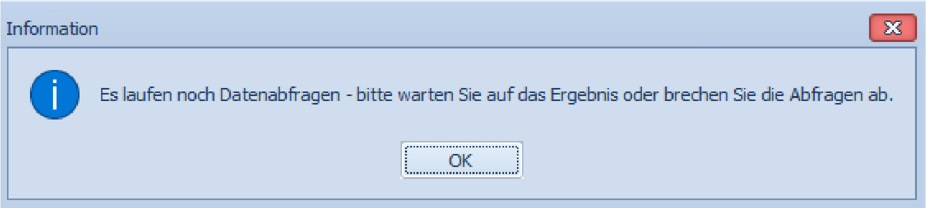
- Operation bar in the planning board (Gantt)
- Operation in the tabular pool of workplaces
- Operation in the tabular pool of groups

Following, you see the order progress for the whole order in the order overview.

Requirement: The option "Update order overview automatically when selecting an operation" is set and the order overview is opened in parallel.

Each click updates the order overview and switches it to the front. Therefore the order overview should be placed on a second screen.

If you click on different operations too quickly in succession, the update may be delayed and a popup will be displayed, indicating that data queries are still running and asking you to wait for the result.



Presentation

Period of the past

The setting *Period of the past* identifies the range/period that is displayed left to the blue line indicating "now".

Time scale

Define the time scale that you want to show on top of the chart in the Graphic order sequencing. Possible values: Minutes, hours, days, weeks, months.

Show operations in pool of groups of the Gantt

If this option is set, the operations included in the pool of groups are shown in the pool of groups of the Graphic order sequencing (Gantt). The operations are shown there at their scheduled date.

If you want to display the operations only in the Gantt pool of groups, you must hide the tabular pool of groups.

One of the two presentations of the pool of groups should always be visible.

Alternative presentation of relationships

Not used.

Save options

Save automatically updated planned dates

If you close the graphic order sequencing or request new data, the prompt that asks if you want to save the changes distinguishes: Are the changes due to explicitly performed planning changes (manual planning, calling the automatic assignment, ...) or are the changes due to the automatic update (moving delayed operations (planned and running) to the line indicating "Now").

- The option "**For planned operations**" is set:

  The prompt that asks if you want to save the planning also appears if operations are implicitly re-planned.

- The option "**For running operations**" is set:

  Running OPs are saved if the planned end date has been recalculated.

1.2.2 Tab *Bar layout*

Bar colors

Color gradient

If the option *Color gradient* is set, the operation bars in the *Graphic order sequencing* are displayed with a color gradient.

3D effect

(reserved)

You can choose from the following options to display the processing time in color:

Status

The operation bars are displayed in the color that has been defined for the respective operation status in the status assignment. This way, currently logged on OPs can be identified via the respective color.

Predecessor status

The operation bars are displayed in the color that has been defined for the status of the predecessor operation in the status assignment.

Order

Operations with the same order number are shown in the same color. The color is selected randomly using internal 8-color iteration.

Order type

Operations with the same order type are shown in the same color. The color is selected randomly using internal 8-color iteration.

Article

Operations processing the same article are displayed in the same color. The color is selected randomly using internal 8-color iteration.

Tool

Operations using the same tool are displayed in the same color. The color is selected randomly using internal 8-color iteration.

Processing time

The bar of the processing time is displayed in the color that has been configured.

Color from OP user field

If a color has been defined for an operation in the configured user field (possible user fields: 7 - 28), the operation is displayed in the respective color. In this case, coloring depends on the number entered in the user field.

Number	Color
1	Green
2	Bright red
3	Pink
4	Purple
5	Black
6	Gray
7	Turquoise
8	Light blue
9	Blue
10	Brown
11	Light green
12	Yellow
13	White
14	Olive
15	Silver
16	Light yellow

Color of running OPs

Running operations can be displayed in a user-defined color.

Display setup times

Enable the option *Display setup times* to show setup times as part of the operation bar. If you have enabled the option, you can display setup time, dynamic setup time or retooling time (teardown time) in one of the available colors.

Setup time

Select the color for the display of the setup time that is defined and stored in the operation.

Teardown/retooling time

Select the color for the display of the teardown time that is stored in the operation.

Operation delayed

If the option *Operation delayed* is enabled, delayed operations are shown in a hatched pattern or in color. An operation is considered delayed if its planned end extends beyond the latest end date (LET/SEZ).

Minimum bar length

If the remaining run time of an operation is less than or equal to 0, the processing time of this operation is shown with the duration entered in the field *Minimum bar length*. Consequently, you can still edit (select, move) the operation.

Note available

If the option "Note available" is enabled, the operation bar shows the icon indicating that a note is available for the operation.

1.2.3 Tab *Tooltip/bar text*

In tab *Tooltip/bar text*, define the information you want to display in the operation's tooltip or in the processing bar.

Tooltip

Click the section *Click here to add new tooltip* to select an additional text to be shown in the tooltip. A selection window opens. Click on the text you want to display in the tooltip. The selection window closes. The text is included in the list.

You can change the order in the tooltip. Click an entry in the list and move it without releasing the button. Release the left mouse button to drop it (drag and drop). The selected entry is moved behind the position where you release the left mouse button.

To delete an entry, select the entry you want to delete and click ✕.

Available tooltip information

The information in the tooltip is read from the database when data is requested.

Some data might change dynamically during planning. In the below table this data is identified by an "X" in the column Dynamic modification.

Description	Dynamic modification
Order	
Sequence	
Operation	
Split	
MES order number	
Operation status	
Secondary status	
OP name	

Description	**Dynamic modification**
Article	
Article designation	
Fixed (yes, no)	
Note available (the internal indicator is shown): X - at least one note is available but no note is identified with the option "display on terminal". T - at least one note is available and at least one note is identified with the option "display on terminal".	
Tool	
Farbe	
Material	
Target cycle	X
Actual cycle	
Target quantity (P) in the primary quantity unit	
Yield (P) in the primary quantity unit	
Remaining quantity (P) (target quantity - yield in the primary quantity unit)	X
Unit (primary quantity unit)	
Partitioning	X
Setup time	X
Dynamic setup time (not relevant in the *Graphic order sequencing*)	
Current setup time (Setup time + dynamic setup time)	X
Target duration (see note below!)	X
Earliest start	
Latest start	
Earliest end	
Latest end	
Scheduled start time	
Scheduled end time	
Planned start date	X
Planned end date	X
Basic start date	
Basic end date	
Priority	
Order index	
First logon	
Last logon	
User fields	

Notes on specific fields:

Target duration:

The field *Target duration* does not display the net processing time according to the remaining run time formula. Instead it shows the bar length shown in the graphic order sequencing and does not account for shift breaks or any other breaks. The processing time depends on the status of the operation:

- For operations that are not logged on (status is not "running"), the bar length equals the sum total of setup time + processing time according to the remaining run time formula + teardown time + dynamic setup time (if available);
- For operations that are logged on, the bar length equals the sum total of the processing time according to the remaining run time formula + retooling time.

The calculation is based on current values (e.g. remaining quantity to be produced). If a field that is included in the calculation of the remaining run time is changed in the database (e.g. because of an update in the PPS system), the changed value is only available once the graphic order sequencing has been reloaded.

User fields: You can select different user fields that either refer to operations or orders. The configuration of the user field keys AGNR/SYSTEM or AUNR/SYSTEM specifies the range of selectable user fields.

Bar legend

Click the section *Click here to add new text* to select an additional text to be shown in the OP bar. A selection window opens. Click on the text you want to display in the OP bar. The selection window closes. The text is included in the list.

To delete an entry, select the entry you want to delete and click ✕.

Available information

Description
Order
Sequence
Operation
Split
MES order number
Operation status
OP name
Article
Article designation
Tool

Description
Target cycle
Priority
Order index

To enable the settings, you must request data in the *Graphic order sequencing*.

1.2.4 *Workplace* tab

Visible columns

The left hand side of the graphic order sequencing shows the capacity groups including the assigned individual capacities (workplaces/machines) that match the planner's responsibility area or user profile.

You can display the following columns in this list of workplaces:

Column	Meaning
Status	Status color or RPA color (depending on the option settings for "Use status text colors")
Workplace	Workplace/machine number
Short name	Short name of the workplace.
Designation	Full name of the workplace
Group	Group of the workplace.
Symbol	Graphic See also the description for the button "Download workplace icons from server".
Performance level	Current performance level of the workplace.

Change the column widths directly in the table (click and drag across). This modification is saved user-specifically.

We recommend to display at least the workplace/machine number.

Use status text colors

If this option is enabled, the workplaces are colored according to the currently active status. The individual status color is defined in the assigned status text. If no color is defined for a status text, the workplace is shown without a color (white/transparent).

If this option is not set, the machines are colored according to the configured RPA.

Show expected malfunction period

When the workplace/machine status is updated, the expected duration of a malfunction entered in the Windows terminal can also be transferred.

If this option is set and if a status with a duration > 0 is available, the expected duration is displayed in the graphic order sequencing. The expected end is calculated:

Expected end = maschinen_status.prog_begin_dat/zeit + maschinen_status.prog_dauer (compared to the Gregorian calendar)

If the expected end is in the future, the period of time between "now" and the expected end is shown as a red bar. The bar does also cover periods without shift (times without shift are not considered). At the same time, a respective symbol is displayed in the column *Symbol* of the workplace list. This symbol overrides the symbol that is normally displayed.

Tooltip

The red bar includes a tooltip showing the following information:

- Workplace: workplace number according to configuration
- Short name: short name of the workplace according to configuration
- Designation: name of the workplace according to configuration
- Current status: status number and status text of the status currently available for the workplace.
- Status since: date, time
- Expected malfunction period: duration in hrs:min:sec.
- Expected end: date, time

Conversion of malfunction period into times without shift

The red bar includes a context menu. The context menu includes the menu item *Convert to time without shift*. Once you have clicked this menu item, the dialog *Individual shift times* opens. The period of time starting from "now" until the expected end is pre-assigned.

Malfunction period does no longer apply

If the malfunction duration returns to 0 the next time the status is identified, the bar and the symbol are removed and the initial symbol is shown again. If this period has been defined as time without shift, this period remains and the planner must remove it explicitly.

Download workplace icons from server

By default, the following symbols are displayed:

- Single workplace according to configuration: 

- Group workplace according to configuration: 

Using this button, the planner can download workplace-specific symbols from the server and store them locally so that they are displayed in the list of workplaces.

For this, the system selects and transfers all files following the scheme *_small.png in the directory defined via the logical Path MOCWPIMG.

When the list of workplaces is loaded, the system checks for every workplace if a file exists in the local directory. If a file exists, the system displays this file (this image). If there is no file, the system checks if the local directory includes a file for the group the workplace is assigned to. If there is a file for the group, this file is displayed.

The search process uses the file name configured for the workplace in the Resource configuration. File sizes should be as small as possible for the display in the list of workplaces. For this reason the file must comply with the naming conventions described below:

Example: File name configured for the workplace: 12330.jpg

Extract the front part of the file name → 12330
Add "_small" →12330_small
Add the file extension ".png" →12330_small.png
Search for file 12330.png
If found, the image 12330_small.png is displayed in the list of workplaces.

Otherwise:

The default image is displayed that references the configured workplace type:

- E = Single workplace: 

- G = Group workplace: 

1.2.5 Tab *Priorities*

Display priorities

You can use the priority of an operation as control tool. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

If this option is enabled, the operation's priority is shown on the left as a colored triangle:

You can define the color for the individual priorities as you like.

Also in the toolbar, you can activate/deactivate the symbol used to display the priority.

Show delays

Using this option, you can identify operations that are expected to be delayed. A narrow bar is displayed on top of the operation bar, if the planned end of a planned operation is x hours earlier than the latest end of the operation (LET). Coloring depends on the configured hours.

1.3 Tabular *Pool of groups*

Go to tab Planning details in the toolbar and click the icon *Pool of groups*.

The tabular pool of groups shows the operations planned for a group. It shows the operations of *all* groups of the graphic order sequencing. The planner can individually define the sorting of the table.

Using the table's filter function, you can restrict the pool of groups to single groups. Refer to the documents on the general operation (moc_cc.pdf) for detailed information on the use of the filter function.

1.4 Tabular *Pool of workplaces*

The tabular pool of workplaces shows all planned operations of a workplace in a tabular list. The list shows operations sorted by their planned start date. However, running operations are always displayed on top of the list, chronologically sorted by logon time. You cannot change the sorting via the table sorting function.

Change to the pool of workplaces of another workplace by single clicking the required workplace in the Gantt workplace table.

The fields that are available in the tabular pool of workplaces are the same as the ones in the tabular pool of groups.

Single clicking an operation in the tabular pool of workplaces

If you use the mouse and single click on an operation with the left button, the system searches for the relevant operation in the other currently active views/dialogs and selects it.

Double clicking an operation in the tabular pool of workplaces

If you double-click (left) an operation, the Order overview is called. When you call the order overview, the current order/operation is transferred. In the order overview, the system automatically displays the order in the upper detail application and the order's operations in the lower detail application (order progress).

1.5 Notes

You can enter notes on an operation. You can view the notes already entered for an operation using the detail application *Notes*, which can be activated from the Toolbar.

The detail application includes a table and a memo field. The table shows the following columns on the left hand side: short text, modified by, date, time, display on terminal, MES order number. To the right, the memo text is displayed. The position of the splitter between table and memo field is saved.

If the detail application *Notes* is active, an additional tab *Notes* is shown in the toolbar. This tab contains the functions needed to edit notes:

Insert a note

You can insert a new note using the operation's context menu. You require the function authorization edopnote.create to insert a new note. The dialog matches the editing dialog available in the order management. When you call the dialog, the MES order number of the operation is pre-assigned. You must enter a short text and a long text.

Disable the checkbox *Display on terminal* if you do *not* want the note to be displayed on the shop floor terminal AIP.

Edit a note

You can edit an existing note using the respective button in the toolbar.

Delete a note

You can delete an existing note using the respective button in the toolbar.

Visualization with the operation bar

If at least one note is assigned to an operation, a symbol  is displayed with the operation bar in the graphic order sequencing (Gantt). If the detail application *Notes* is visible and you click one or more OP bars, the contents of the detail application *Notes* are updated.

1.6 Planning functions

1.6.1 Manual scheduling using drag & drop

Usually, the operations are planned for a (capacity) group in the ERP system. Once the individual capacities (workplaces) are loaded into the planning board, you use drag & drop to schedule operations from the tabular or graphic pool of groups for the workplace. Left-click an operation and drag it to the required location in the graphic planning board.

1.6.2 Replan an operation for another group

It is generally possible to re-plan an operation for another capacity group or a workplace of another group in the graphic planning board. But this is **not** a planning function in the sense of the above definition. Note the following:

- Process times are not recalculated during replanning.
- Assigned resources (components, production resources and tools) remain assigned.
- The application does not check if replanning makes sense (e.g. you replan a drilling operation for a "milling" group).

1.6.3 Deallocate operation

If you deallocate an operation, the operation is returned to the pool of groups. Depending on the setting of the option "Show OPs in the pool of groups of the Gantt", proceed as follows:

Option "Show OPs in the pool of groups of the Gantt" is set

To **deallocate** an operation, use drag & drop to select the required operation in the planning board and move it back into the *graphic* pool of groups in the planning board.

The operation is displayed at the scheduled start time in the pool of groups of the planning board (not necessarily the same place where you "dropped" the operation).

Option "Show OPs in the pool of groups of the Gantt" is not set

To **deallocate** (unplan) an operation, select the required operation in the planning board and move (drag and drop) it into the graphic *pool of groups in the planning board* (not into the tabular pool of groups).

When you save planning, the system sets the option "planned" to "group" for deallocated operations. The workplace the operation was previously planned for is NOT deleted.